

Synch Write Asynch Read RAM

Parameters

- DATA_WIDTH.
- OUTPUT_SIZE.
- MEM_SIZE = 512.
- ADDR_SIZE = 9.
- DATA_SIZE = DATA_WIDTH * OUTPUT_SIZE.

[DATA_SIZE – 1 : 0] mem [MEM_SIZE – 1 : 0]

Input Signals

- rst_n -> 1 bit -> asynchronous reset put every output = 0.
- out_read_en -> 1 bit -> when high, the out_read_data = the data in the address out_read_address.
- out_read_address -> ADDR_SIZE bit -> contains the address for the data outputted to the out port.
- alu_read_en -> 1 bit -> when high, the alu_read_data = the data in the address alu_read_address.
- alu_read_address -> ADDR_SIZE bit -> contains the address for the data outputted to the alu port.
- write_en -> 1 bit -> when high

Signal	Number of bits	Description
rst_n	1	Asynch reset put every output = 0
out_read_en	1	when high, the out_read_data = the data in the address out_read_address else = 0
out_read_address	ADDR_SIZE	contains the address for the data outputted to the out port
alu_read_en	1	when high, the alu_read_data = the data in the address alu_read_address else = 0

alu_read_address	ADDR_SIZE	contains the address for the data outputed to the alu port
write_en	1	When high it makes the data on data_in written in write_address
write_address	ADDR_SIZE	The address which the data_in is written in
data_in	DATA_SIZE	The data written in the write_address

Output Signals

Signal	Number of bits	Description
out_read_data	DATA_SIZE	Data in the out_read_address if out_read_en is high else = 0
alu_read_data	DATA_SIZE	Data in the alu_read_address if alu_read_en is high else = 0